

University Question Bank

Unit - I

- 1) Construct NFA equivalent to the Regular Expression: $(0+1)^*01$.
2. Distinguish NFA & DFA.
3. Define DFA.
4. Give the purpose of Transition diagrams.
5. List out the applications of Automata theory.
6. What is Regular language?
7. Define ϵ -closure property.
8. Differentiate b/w Moore & Mealy Machines.
9. What is Regular Expression?
10. Construct two finite automata for the Regular Expression $(0+1)^*(0+1)^*$.

11. Write Regular Expression for a set of strings over alphabet $\{a, b, c\}$ containing at least one a & at least one c .



Wk 16 • 107-258
Wednesday
April

Unit-II

8.00

9.00 1. List the closure properties of Regular language

10.00 2. When a grammar is in CNF?

11.00 3. When is a grammar Ambiguous? Give Example

12.00 4. Define GNF.

1.00 5. What is a Context free grammar?

2.00 6. Define Induction principle

3.00 7. What is the closure property of Regular sets?

4.00 8. What is Homomorphism?

5.00 9. List any two applications of Regular Expression.

6.00 10. State the Relations among Regular Expression, deterministic finite automata, NFA, GFA with ϵ -transitions.

11. find out the CFL $S \rightarrow asb | aab; A \rightarrow bAa$
 a

Notes

12. Define left most derivation.

13. What is a production?

14. What is a derivation tree? Give Example.

Unit - III

What is a pushdown automata?

What is Recursively Enumerable Language?

Does the CFL have an equivalent Deterministic or Non-Deterministic pushdown automata for the same language?

State the pumping Lemma for context free languages.

Define Sentential form.

What is Deterministic pushdown automata.

Write the PDA for empty stack?

How can we prove that a given language is not a CFL?

What is Bottom up parsing?



Wk 16 • 109-256
Friday
April

10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

Unit - IV

8.00

9.00 1. What is Recursive Language?

10.00 2. Compare Total & partial Recursive function.

11.00 3. What is a Turing Machine?

12.00 4. What is a primitive Recursive fn?

1.00 5. Define Recursive Enumerable Language.

2.00 6. What is the Storage in FC?

3.00 7. Does a ~~TM~~ NTM accept a language, accepted by TM (or) DTM. Justify.

4.00 8. Give the instantaneous description of TM.

5.00 9. State the properties of that satisfy recursively enumerable language.

6.00 10. Define Multitack Machines.

7.00 11. What is offline Turing Machine.

Unit - 2

- 1) What is Space Complexity?
- 2) What do you mean by Conjunctive Normal form?
- 3) What is NP Complete class?
- 4) Define Exponential form & polynomial time Reduction.
- 5) When do you say a problem is undecidable? Mention any two undecidable problems.
- 6) Is the travelling Salesman problem NP complete? If so prove.
- 7) What is meant by Time Complexity?
- 8) What is meant by NP class?
- 9) What is Running time of Turing Machine?
- 10) Which problems belongs to class P & NP?
- 11) Compare time & Space Complexity.
- 12) State the halting problem of TM.
- 13) What are NP class?
- 14) State & Satisfiability problem?

Wk 17 • 111-254

Sunday
April



8.00

Unit 29.00 1) Equivalence of NFA with @ without ϵ - How.10.00 2) If there is a language L , such that $L = L(A)$ for some DFA A , then there is a Regular Expression R such that $L = L(R)$.

11.00

12.00 3) $(0+1)^* (0+1)$

1.00 4) Minimization of DFA.

2.00 5) Conversion of NFA to DFA.

6) Moore & Mealy M/c.

3.00

Unit - II

4.00 1) properties of Regular sets in Detail.

5.00 2) $S \rightarrow SS / 0S1 / 01$ into CNF with theorem6.00 3) $S \rightarrow aSA / aAA / b$
 $A \rightarrow bBBB$

7.00

~~B~~ $B \rightarrow b/c$ into CNF with theorem

4) parse trees

Notes

Appointments

Phones

Unit - III

- 1) properties of CFL with Example.
- 2) pumping lemma for CFL.
- 3) If L is context free language, then prove that there exists a PDA M such that $L = N(M)$.
- 4) push down Automata
- 5) Decision Algorithms

Unit - IV

- 1) $L = \{a^n b^n a^n b^n : n \geq 1\}$ and for binary language for palindrome strings.
- 2) Various Types of Turing M/c.
- 3) Recursive Enumerable Languages.
- 4) Undecidable pblms

Unit - V

1) Time & space Complexity of TM

2) NP problems

Vertex cover

SAT

Clique

Appointments

Phones

3) NP Hardness & NP

Completeness. Also explain when a pblm is NP-hard & when it is NP-complete